



General Sir John Kotelawala Defence University, Sri Lanka

Faculty of Engineering

**Department of Civil Engineering**

# **Vertical Axis Wind Turbine**

**FINAL REPORT**



**Fluid Mechanics - Assignment 01**

## **Module**

CE 1232 - Fluid Mechanics

Assignment 01

Dimensional Analysis

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# AGENDA

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2. Objectives
3. Project Execution
4. Dimensional Analysis
5. Observation & Results
6. Discussion
7. Questions & Answers
8. Conclusion
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# INTRODUCTION

This project focuses on developing a **vertical axis wind turbine (VAWT)** with helical blades, demonstrating its feasibility for energy generation in urban and rural settings. VAWTs offer advantages like omnidirectional wind capture and simpler mechanics compared to horizontal axis turbines. The project aimed to construct a prototype model and analyze its performance using scaled measurements.

In today's era of very serious pollution, people can be said to be thoughtful to deal with environmental problems. Today the topic of wind power generation is attracting more attentions. First, wind power generation is an important part of developing new energy and improving the environment. In addition, compared with coal-fired power plants of the same scale, wind farms emit zero pollutants into the atmosphere, achieving zero solid and gas emissions.

Wind power is one of the main directions of future renewable energy development. Wind turbines can be classified according to their axes. The two major categories are vertical axis wind turbines and horizontal axis wind turbines, both types have their unique characteristics in efficiency, operating principle, and operating environment.

Structure of Vertical Axis Wind Turbine — The vertical axis wind turbine is mainly composed of wind turbine blades, main rotor shafts, power generation components, and fixed bases. Its main axis of rotation is mounted vertically to the wind direction, so there is no need for a pitch mechanism to adapt to the wind direction. The blades are mounted on the main rotor shaft. The spindle is connected to the power generation module at the base. The power generation component consists of a gearbox and a generator. Vertical axis wind turbines can be classified into two types: resistance type and lift type according to blade design. Among them, the resistance blade design uses the resistance generated by air passing through the blade to drive the rotation of the main rotor shaft, and the lift blade design uses the lift generated when air flows through the blade to drive the rotation of the main rotor shaft. When the blade rotation speed is faster, the resistance generated by air passing through the blade decreases, and the lift force increases.

Generating Mechanism of Vertical Axis Wind Turbine — The vertical wind turbine blades convert the kinetic energy of the wind into which the kinetic energy of the main rotor shaft is converted. The spindle is connected to the gearbox to adjust the torque of its output rotational motion and its rotational speed to match the generator for higher power generation efficiency.

Characteristics of Vertical Axis Wind Turbine Vertical-axis wind turbines do not need to counter the wind when the wind direction changes, which is a major advantage over horizontal-axis wind turbines. It not only simplifies the structural design but also reduces the torque of the wind wheel against the wind. Breeze starting (1m/s can start power generation), but the tip speed ratio is low, and the power output is low when the size, weight, and cost of the wind wheel are certain.

## OBJECTIVES

1. Construct a scaled model of VAWT.
2. Select appropriate materials to simulate real-world turbine performance.
3. Apply **dimensional analysis** to ensure the prototype reflects real turbine dynamics.
4. Identify challenges while constructing VAWT.

# PROJECT EXECUTION

## **Design and Materials Used**

- **Design Elements:**
  - **Helical Blades:** Smoother rotation and reduced torque fluctuations.
  - **Vertical Axis:** Captures wind from all directions
  
- **Materials:**
  - **Blades:** Sheet Metal
  - **Tower:** PVC
  - **Arms and Base:** Metal and Plastic
  - **Fasteners:** Nuts, bolts, bearing, glue
  
- **Tools:** Glue gun, saw, screwdrivers, spanners, scissors, and rulers.

## **Assembly Process**

1. **Blade Construction:** Shaped metal sheet helical blades according to the scaled design.
2. **Tower Assembly:** Built the tower using PVC pipes.
3. **Rotor Assembly:** Attached the blades to the vertical axis.
4. **Final Assembly:** Blade and tower are connected with bearing which allows to helical blade to rotate freely about the axis of the tower

# DIMENSIONAL ANALYSIS

To ensure that the prototype correctly reflects the behaviours of a full-scale wind turbine, **geometric analysis** and similarity principles were applied.

|                                      |
|--------------------------------------|
| Scale ratio   prototype: Model = 3:1 |
|--------------------------------------|

|                  | MODEL              | PROTOTYPE |
|------------------|--------------------|-----------|
| Number of blades | 3                  | 3         |
| Helical Angle    | 58.36 <sup>0</sup> | 58.36°    |
| Rotor Height     | 510mm              | 1530 mm   |
| Rotor Diameter   | 300mm              | 900 mm    |

The prototype dimensions were provided by, **Mr. MM Kothalawala** whose contribution was invaluable to the design process. The prototype dimensions used in this project were adopted from the research conducted by Kothalawala et al. (2024) on “Design and analysis of rotor and airfoil configurations for vertical axis wind turbines on highways”.

## Results

- Specifications of the wind turbine for the optimum efficiency.

|                           |          |
|---------------------------|----------|
| $\lambda$ - Optimum       | 4        |
| Power coefficient optimum | 0.4560   |
| Maximum mechanical power  | 130.28 W |
| Cut in speed              | 2m/s     |
| Rated speed               | 7m/s     |
| Cut out speed             | 11.78m/s |
| Chord Length              | 100mm    |
| Number of blades          | 3        |
| Helical Angle             | 58.36°   |
| Rotor height              | 1530mm   |
| Rotor Diameter            | 900mm    |
| Solidity                  | 0.333    |

27/09/2024

Footer

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## OBSERVATION & RESULTS

- **Performance:**

- The turbine rotated smoothly at low wind speeds.
- The helical design provided stable rotation with minimal torque fluctuations.
- Power output was limited due to the small size and low tip-speed ratio typical of VAWTs.

- **Challenges:**

- Aligning blades from the correct angle was a challenge, leading to multiple adjustments.
- We faced difficulties welding the metal rod with the bearing
- We must be aware while welding because slugs may get into the bearing it may affect their smooth rotation.
- In here we faced the challenges when dealing sharp metal sheet which is use for helical blade.
- Material limitations affected the prototype's durability, suggesting the need for composites in future iterations.

## DISCUSSION

The design and construction of the Vertical Axis Wind Turbine (VAWT) prototype offered valuable insights into the challenges and advantages of this turbine type. The use of helical blades was crucial in achieving smoother rotation. This feature ensures stable operation in low and turbulent wind conditions, making VAWTs suitable for urban settings where wind direction is highly variable.

## QUESTIONS & ANSWERS

**1. State your practical phenomenon, identify the parameters involved and derive the nondimensional groups involved in your application.**

The Vertical Axis Wind Turbine (VAWT) converts kinetic energy from wind into rotational energy to generate power. The goal of dimensional analysis here is to ensure that the scaled prototype reflects the performance and behaviour of a full-scale turbine by using appropriate non-dimensional parameters.

The physical parameters influencing the behavior of VAWT include:

- **V:** Wind speed (m/s)
- **D:** Characteristic length, typically blade width or turbine diameter (m)
- **$\rho$ :** Air density (kg/m<sup>3</sup>)
- **$\mu$ :** Dynamic viscosity of air (kg/m·s)
- **P:** Power output (W)
- **A:** Swept area of the blades (m<sup>2</sup>)
- **$\omega$ :** Angular velocity of the rotor (rad/s)

Non-Dimensional Groups

- a. Reynolds Number:  $Re$  is dimensionless and helps determine whether the air flow is laminar or turbulent.
- b. Power Coefficient: It helps compare the efficiency of different turbine designs.

## **2. What changes would you consider in the model making process, material selection and testing in order to achieve complete similarity?**

For achieving complete similarity between the prototype and the actual model, you'll need to address not only geometric similarity but also dynamic similarity through appropriate material selection, model making, and testing adjustments. Below is a detailed answer tailored to your project's VAWT prototype (3:1 scaling ratio) and focused on dimensional analysis principles.

- Geometric Similarity: Matching shapes and proportional dimensions.
- Kinematic Similarity: Ensuring velocities and accelerations are proportional.
- Dynamic Similarity: Matching forces, such as inertial, viscous, and gravitational forces, between the model and prototype.

Material Selection: Use Lightweight Materials with Similar Mechanical Properties like aluminium, carbon Fiber

## **3. Will you be able to achieve complete similarity in the application chosen? Explain.**

We just only consider about geometric similarity only because if we consider complete similarity we have to consider about dynamic and kinematic similarities for achieving those things we more technical knowledge on (CFD) computation fluid dynamics and other related software and also we need more laboratory equipment to analysis dynamic and kinematics similarities

## CONCLUSION

This project successfully demonstrated the design, construction, and testing of a scaled **Vertical Axis Wind Turbine (VAWT)** with helical blades. The use of **dimensional analysis** ensured that the scaled prototype maintained geometric similarity with a full-scale turbine, enabling us to derive meaningful insights into its appearance. The helical design allowed for smooth and omnidirectional operation, showcasing the potential of VAWTs for urban and rural applications where wind direction is variable.

While the prototype met the fundamental design objectives, challenges were encountered during construction, particularly with material selection and blade alignment, which affected rotational stability. The project emphasized the importance of material optimization and precise assembly techniques for future improvements.

Incorporating the dimensions provided by **Mr. M. M. Kothalawala** from his research on **rotor and airfoil configurations** was crucial in achieving accurate scaling. This collaborative effort not only enhanced the practical aspect of the project but also highlighted the value of leveraging existing research to inform design processes.

Moving forward, further testing under controlled conditions, such as **wind tunnel experiments**, will provide more accurate performance data. Additionally, using **lightweight composites** and exploring **alternative blade designs** could enhance the turbine's efficiency. This project provided valuable hands-on experience in engineering design, offering insights into the challenges and potential of **renewable energy technologies**.

## RECOMMENDATIONS

1. Use **lightweight composites** to improve efficiency.
2. Conduct **wind tunnel tests** to obtain more accurate data.
3. Scale up the model for further testing to better simulate real-world conditions.

## REFERENCES

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